Internship Report: AWS in a Smart Parking Management System

Introduction

During today's internship session, I learned about Amazon Web Services (AWS) and its practical applications in cloud computing. Instead of studying only the individual AWS services, I wanted to understand how they can be combined to solve a real-world problem. I selected a Smart Parking Management System because finding a parking space in shopping malls, hospitals, airports, offices, and public places is often difficult and time-consuming. A cloud-based parking system allows users to check parking availability, reserve a slot, receive navigation assistance, and make online payments. AWS provides the infrastructure required to build such a system in a secure, reliable, and scalable way.

Why Cloud Computing?

If an organization hosts the parking application on its own servers, it must purchase hardware, maintain storage, install software updates, perform backups, and ensure security. As the number of users grows, upgrading the infrastructure becomes expensive. By using AWS, these responsibilities are managed by the cloud provider. Resources can be increased or decreased whenever required, making the application cost-effective and capable of handling changing traffic without major maintenance.

Amazon EC2

Amazon EC2 can host the main Smart Parking web application or mobile backend. Users can create accounts, search nearby parking locations, reserve available spaces, make payments, and check booking history through the application. During weekends, festivals, or office hours, a large number of users may access the system simultaneously. Additional EC2 instances can be launched whenever demand increases, ensuring that the application continues to perform smoothly.

Amazon S3

The parking system may store images captured by CCTV cameras, parking entry and exit photographs, payment receipts, invoices, and user-uploaded documents. Amazon S3 provides secure, durable, and highly available storage for these files. Storing files in S3 instead of on the application server improves storage management and reduces the load on the server. Older files can also be archived using lower-cost storage classes.

Database Services

Important information such as user profiles, vehicle registration numbers, parking slot availability, reservation history, payment details, and transaction records can be stored in Amazon RDS. Since RDS is a managed database service, backups, software updates, and database maintenance are handled automatically. For applications requiring very fast access to frequently changing parking availability, Amazon DynamoDB can also be used.

AWS Lambda

AWS Lambda can automate several background tasks without keeping a server running continuously. After a user books a parking slot, Lambda can automatically send a confirmation email or notification. It can also update slot availability whenever a vehicle enters or exits the parking area, generate digital receipts after successful payment, and remind users before their reservation expires.

IAM and Security

AWS Identity and Access Management (IAM) helps secure the application by providing different permission levels. Customers can manage only their own bookings and payment history. Parking staff can update slot status and verify vehicle entry, while administrators can manage the complete system, generate reports, and control user access. Proper permission management protects sensitive customer information and prevents unauthorized access.

Load Balancer and Reliability

During peak hours, thousands of users may search for parking spaces at the same time. An Elastic Load Balancer distributes requests across multiple EC2 instances, reducing server overload and improving response time. AWS monitoring services help identify performance issues, while automatic backups improve reliability and reduce the risk of data loss.

Conclusion

From today's internship session, I understood that AWS offers a complete cloud platform for building practical applications like a Smart Parking Management System. Services such as EC2, S3, RDS, Lambda, IAM, and Load Balancer work together to create a secure, scalable, and efficient solution. This example helped me understand how different AWS services solve real-world problems instead of only learning their definitions. Overall, this activity improved my understanding of cloud computing and showed how AWS can be applied in everyday applications.